21ASC305T
ROCKET PROPULSION

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- ***Unit-1 - Basic Concepts 9 Hour***
- *Rocket propulsion systems -working principle, Classification of rockets, Rocket equation, Rocket Nozzles and its classifications -Nozzle performance- equilibrium and frozen flow, Characteristic velocity and Thrust coefficient, Performance parameters and efficiencies of rocket, Staging and clustering of rocket*
- ***Unit-2 - Chemical Propellants 9 Hour***
- *Chemical propellants -Molecular mass, Specific heat capacities, Specific heat ratio, mixture ratio, Stoichiometric ratio and equivalence ratio, Energy release during combustion-Heat of formation and Heat of combustion-Combustion Instabilities-Criterion for choice of propellant, Solid Propellants-Composition and processing, Liquid Propellants-Classifications of liquid propellant, Storability of liquid propellant, Chemical Equilibrium*
- ***Unit-3 - Solid Rocket Engine 9 Hour***
- *Propulsion Elements for Solid Rocket Motors, Solid propellant combustion and Internal Ballistics of Motors - Mechanism of burning, Ignition System-Igniter types - Solid propellant grain design - Burn Rate-Factors influencing burn rates, burn rate index for stable operation, Action time and burn time, Design of Solid Propellant rocket, Simulation of solid rocket motor performance.*
- ***Unit-4 - Liquid Rocket Engine 9 Hour***
- *Liquid Propellant Rocket-Hardware components and its functions, Propellant feed systems –Gas pressure feed system, Turbo pump feed systems and engine cycles, Tank Pressurization - Types of injectors – Thrust chamber, Cooling of Thrust chamber - Cryogenic propulsion system and its special features, Simulation of Liquid rocket performance, Hybrid rockets- Review of solid-fuel regression rate behavior in classical and non-classical Hybrid Rocket Motors - Rocket engine design using software.*
- ***Unit-5 - Non-conventional Propulsion Techniques 9 Hour***
- *Non - conventional propulsion techniques -Electrical Rockets-Electro-thermal propulsion system-Arc-jet thruster, Resistojet Thruster-Electro-static thruster-Ion thruster, Bombardment Ionization, Hall Effect Thruster -Electro-magnetic propulsion system-Magneto plasma dynamic accelerator, Pulsed Plasma accelerator- Laser Propulsion -Nuclear powered rockets- Solar Propulsion system*

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- *1. George P. Sutton, Oscar Biblarz, "Rocket propulsion elements", Wiley India Pvt Ltd. Ninth Edition, 2017*
- *2. Ramamurthi.K, "Rocket propulsion", Laxmi Publications, India, Second edition, 2019.*
- *3. J D. Mattingly and K M. Boyer,"Elements of Propulsion: Gas Turbines and Rockets", Second Edition,2016*
- *4. Philip Hill and Carl Peterson, "Mechanics and thermodynamics of propulsion", Pearson India, second edition, 2010.*
- *5. Stephen R. Turns, "An Introduction to Combustion: Concepts", McGraw-Hill Education; 4th edition, 2020*
- *6. Yahya S M, "Gas Tables",8th Edition, 2018*

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Unit-3 - Solid Rocket Engine

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PROBLEM

- An end-burning rocket uses a cylindrical double-base propellant grain with a diameter of 200 mm and generates a thrust of 350 N over a period of 300 s. The thrust coefficient is 1.15. The characteristics of the propellant are:
- Density of propellant grain = 1500 kg/m^3
- a = 4 mm/s; n = 0.5
- C* =1500 m/s
- Determine: (i) the length of the grain
- (ii) throat diameter of the nozzle

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GIVEN DATA

- Grain diameter, $D = 200$ mm $= 0.2$ m
- Thrust, $F = 350$ N
- Burning time, $t_b = 300$ s
- Thrust coefficient, $C_F = 1.15$
- Propellant density, $\rho_p = 1500$ kg/m^3
- Burning-rate coefficient, $a = 4$ mm/s $= 0.004$ m/s
- Pressure exponent, $n = 0.5$
- Characteristic velocity, C∗=1500 m/s

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Propellant mass flow rate

$$\dot{m} = \frac{F}{C_F C^*}$$

$$\dot{m} = \frac{350}{1.15(1500)}$$

$$\boxed{\dot{m} = 0.2029 \text{ kg/s}}$$

Total propellant consumed

$$m_p = \dot{m} t_b$$

$$m_p = 0.2029(300) = 60.87 \text{ kg}$$

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Length of the grain:

For an end-burning cylindrical grain, the **burning area** is the circular end area:

$$A_b = \frac{\pi D^2}{4}$$

$$A_b = \frac{\pi (0.2)^2}{4} = 0.031416 \text{ m}^2$$

The grain volume is

$$V_p = A_b L$$

$$m_p = \rho_p A_b L$$

$$L = \frac{m_p}{\rho_p A_b}$$

$$L = \frac{60.87}{1500(0.031416)}$$

$$\boxed{L \approx 1.29 \text{ m}}$$

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Using burning rate, the length of grain is

$$\dot{m} = \rho_p A_b r_b$$

$$r_b = \frac{\dot{m}}{\rho_p A_b}$$

$$r_b = \frac{0.2029}{1500(0.031416)} = 0.004307 \text{ m/s}$$

$$\boxed{r_b = 4.307 \text{ mm/s}}$$

$$t_b = \frac{L}{r_b}$$

$$L = r_b t_b = (0.004307)(300)$$

$$\boxed{L = 1.292 \text{ m}}$$

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Throat diameter

Chamber pressure using the burning-rate law

$$r_b = a p_c^n$$

$$4.307 = 4 p_c^{0.5}$$

$$\sqrt{p_c} = \frac{4.307}{4} = 1.0768$$

$$p_c = (1.0768)^2$$

$$\boxed{p_c = 1.159 \text{ MPa}}$$

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$$F = C_F p_c A_t$$

$$A_t = \frac{F}{C_F p_c}$$

$$A_t = \frac{350}{1.15(1.159 \times 10^6)}$$

$$A_t = 2.628 \times 10^{-4}\ \text{m}^2$$

$$A_t = \frac{\pi d_t^2}{4}$$

$$d_t = \sqrt{\frac{4A_t}{\pi}}$$

$$d_t = \sqrt{\frac{4(2.628 \times 10^{-4})}{\pi}}$$

$$\boxed{d_t \approx 0.0183\ \text{m}}$$

$$\boxed{d_t \approx 18.3\ \text{mm}}$$

Problem

- **An end-burning solid propellant rocket, having a throat diameter of 28 mm, is employed as the sustainer of a missile. It needs to provide a constant thrust of 5 KN. If a double-base propellant with a burn rate law $r = 0.004P^{0.45}$ where P is in Pa and r is in mm/s is used, determine the diameter of the propellant grain.** You can assume the specific impulse of the booster to be 1900 N.s/kg. The C^* and density of the double-base propellant are 1600 m/s and 1540 kg/m^3 respectively.

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Chamber pressure:

Mass flow rate

$$\dot{m} = \frac{F}{I_{sp}}$$

$$\dot{m} = \frac{5000}{1900}$$

$$\boxed{\dot{m} = 2.632 \text{ kg/s}}$$

$$C^* = \frac{p_c A_t}{\dot{m}}$$

$$p_c = \frac{\dot{m} C^*}{A_t}$$

$$A_t = \frac{\pi (0.028)^2}{4} = 6.158 \times 10^{-4} \text{ m}^2$$

$$p_c = \frac{(2.632)(1600)}{6.158 \times 10^{-4}}$$

$$\boxed{p_c = 6.835 \times 10^6 \text{ Pa}}$$

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Burning rate

$$r = 0.004p^{0.45}$$

$$r = 0.004(6.835 \times 10^6)^{0.45}$$

$$\boxed{r = 8.90 \text{ mm/s}}$$

$$r = 0.00890 \text{ m/s}$$

$$\dot{m} = \rho_p A_b r$$

$$A_b = \frac{\dot{m}}{\rho_p r}$$

$$\rho_p = 1540 \text{ kg/m}^3$$

$$A_b = \frac{2.632}{1540(0.00890)}$$

$$\boxed{A_b = 0.1921 \text{ m}^2}$$

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Propellant grain diameter

$$A_b = \frac{\pi D_p^2}{4}$$

$$D_p = \sqrt{\frac{4A_b}{\pi}}$$

$$D_p = \sqrt{\frac{4(0.1921)}{\pi}}$$

$$D_p = 0.4946 \text{ m}$$

$$\boxed{D_p \approx 495 \text{ mm}}$$

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MODYNAMICS

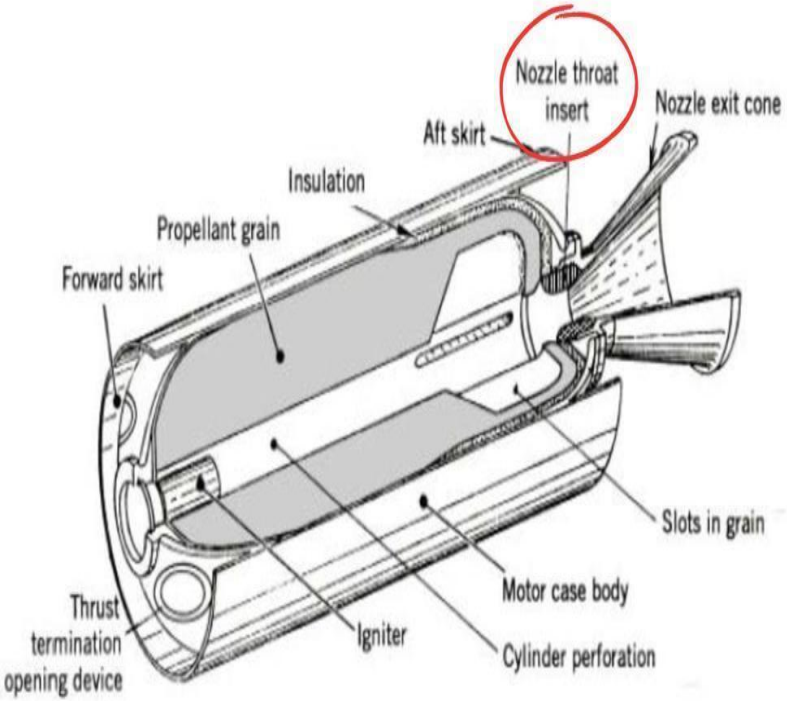
IGNITION PROCESS

- Solid propellant ignition comprises a series of complicated rapid events that start upon the receipt of a signal (usually electric) that ignites the propellant and this hot gas flows from the igniter to the motor grain surface,
- spreading of a flame over the entire burning surface area,
- filling the chamber cavity with gases, and elevating the chamber pressure without serious abnormalities such as
  - overpressures,
  - combustion oscillations,
  - damaging shock waves,
  - hangfires [delayed ignition],
  - flame extinguishment, and chuffing.

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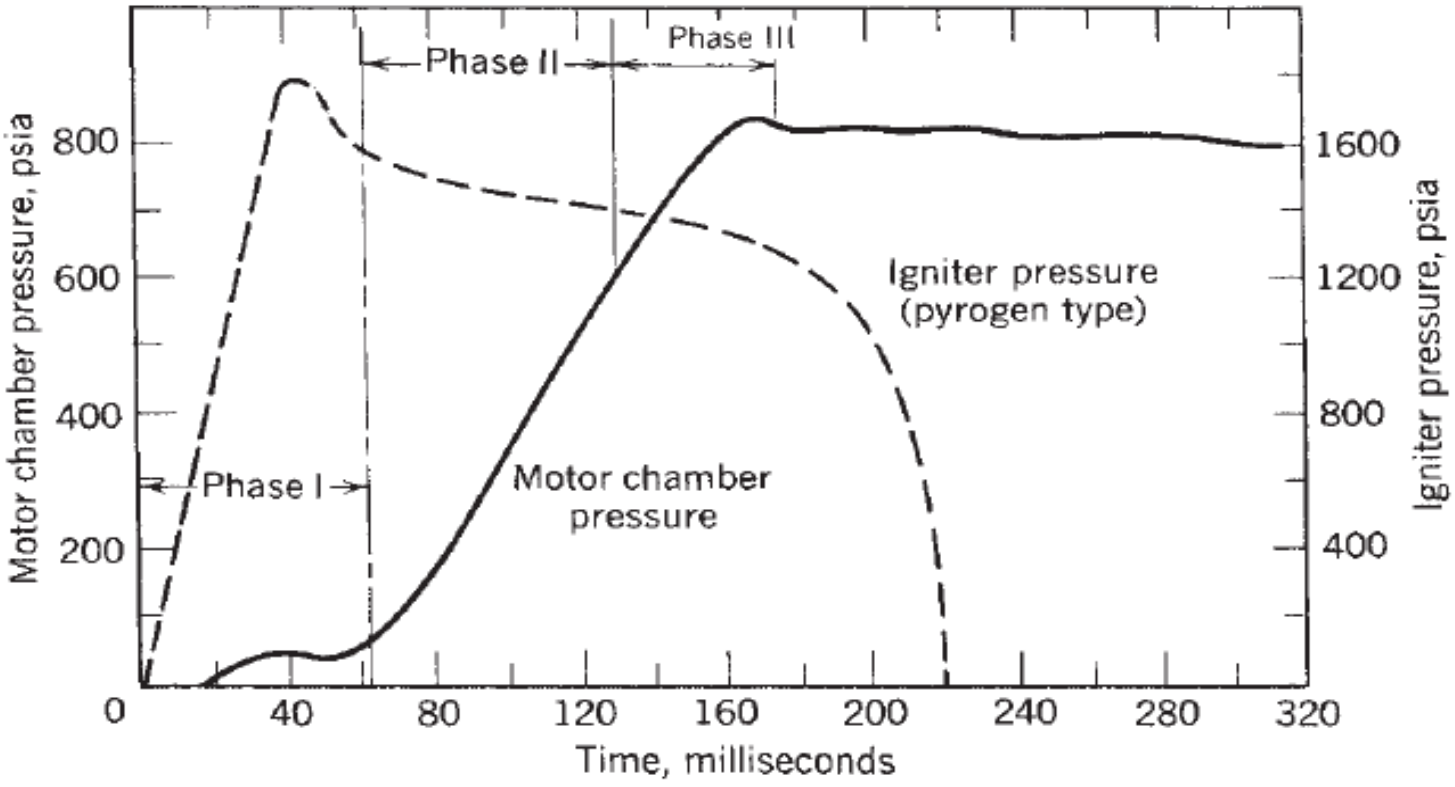
IGNITION PROCESS

- The *igniter* in a solid rocket motor is the component that heats and generates the gases necessary for motor ignition.
- Rocket motor ignition must usually be completed in fractions of a second for all but the very large motors.
- The pressure inside the motor rises to its equilibrium state in a very short time



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IGNITION PROCESS



IGNITION PROCESS

- Conventionally, the ignition process is divided into three phases for analytical purposes:

- **Phase I, *Ignition time lag*:** The period from the instant the igniter receives its signal until a portion of the grain surface burns and produces hot gases.

- **Phase II, *Flame-spreading interval*:** The time from first ignition of the grain surface until the complete grain burning area has been ignited.

- **Phase III, *Chamber-filling interval*:** The time for completing the chamber gasfilling process and for reaching equilibrium chamber pressure and mass flow.

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IGNITION PROCESS

- Ignition will be successful once sufficient grain surface is ignited and burning, so that the rocket motor will continue to raise its own pressure up to the operating chamber pressure.

- If the **igniter is not powerful enough**, some grain surfaces may burn only for a short time and the flame will be extinguished;

- the relevant critical processes seem to be located in a gas-phase reaction above the burning surface, where propellant vapors and/or decomposition products interact with each other and with the igniter gas products.

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IGNITION PROCESS

- Satisfactory attainment of equilibrium chamber pressures with full gas flows depends on

- (1) the characteristics of the igniter and its gas temperature, composition and issuing flow,

- (2) the motor propellant composition and grain surface ignitability,

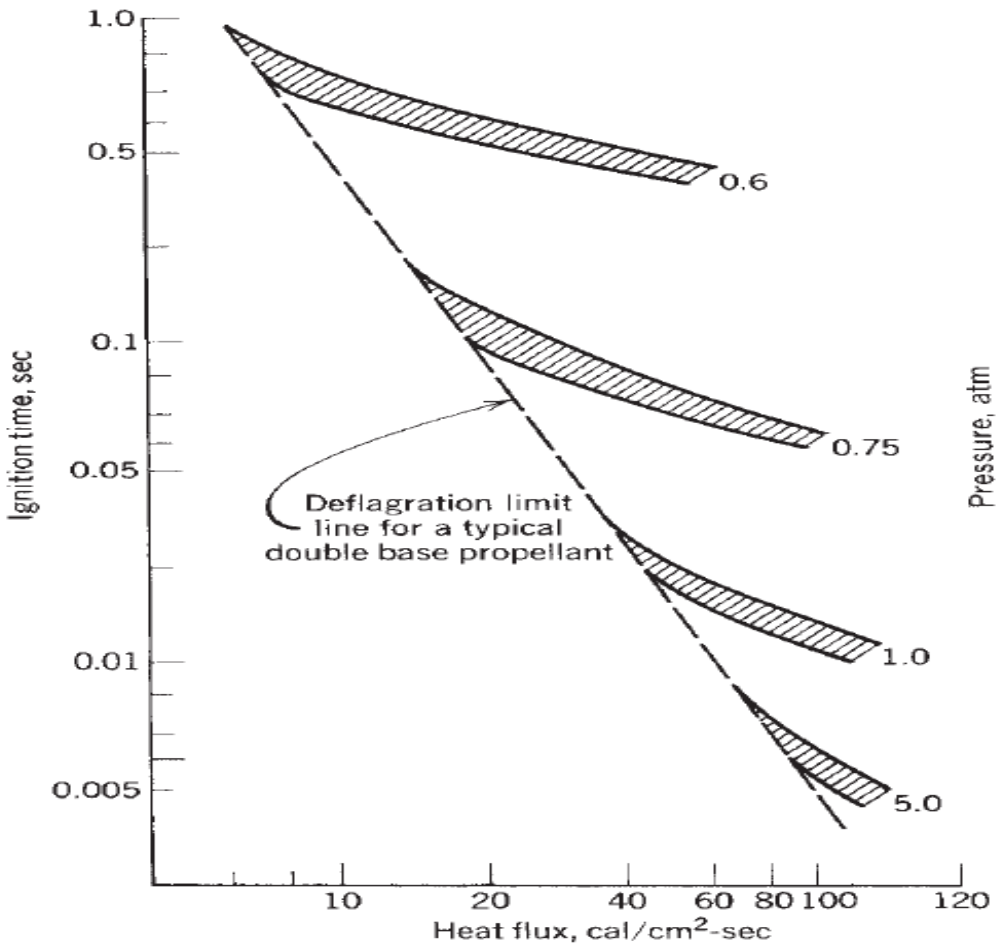
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IGNITION PROCESS

- (3) the radiation and convection heat transfer rates between the igniter gas and grain surface,
- (4) the grain flame spreading rate, and
- (5) the dynamics of filling the motor free volume with hot gas
- The **quantity and type of caloric energy** needed to ignite a particular motor grain in its prevailing environment has a direct bearing on most of the igniters' design parameters—particularly those affecting its required heat output.

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Propellant ignitability curves: effect of heat flux on ignition time



IGNITION PROCESS

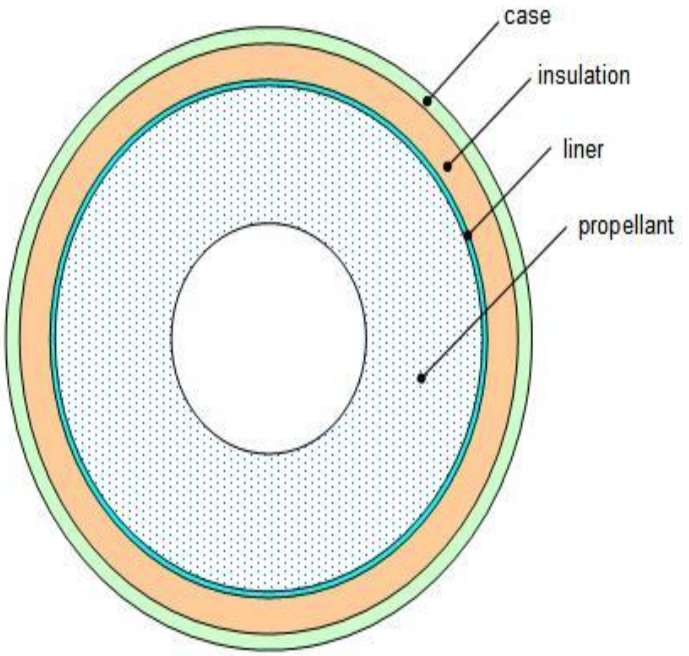
- Propellant ignitability is affected by several factors, including
- (1) propellant formulation,
- (2) propellant grain surface initial temperature,
- (3) surrounding pressure,
- (4) modes of heat transfer,
- (5) grain surface roughness,



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IGNITION PROCESS

- Propellant ignitability is affected by several factors, including
  - (6) propellant age,
  - (7) composition and hot-solid-particle content of igniter gases,
  - (8) igniter propellant and its initial temperature,
  - (9) velocity of hot igniter gases relative to the grain surface, and
  - (10) cavity volume and configuration.



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